DESIGN AND DEVELOPMENT OF ESP32

PROJECT NAME:	Design And Development OF ESP32
PROJECT GROUP:	Sahil Khandare

WHY THIS PROJECT ?

- o learn the process of designing a custom PCB for ESP32.
- To understand the connections and working of the ESP32 microcontroller.
- To gain practical knowledge of circuit design and PCB layout.
- To develop a compact and reliable ESP32 development board.
- To use the board for future embedded and IoT projects.

INTRODUCTION

- This project focuses on the design and development of a custom ESP32-based PCB.
- The PCB was designed to provide proper connections for power supply, programming, and input/output pins of the ESP32.
- Components were mounted on the board to make a compact and functional system.
- The project helped in understanding PCB design, routing, and hardware development.
- The developed board can be used for embedded systems and IoT applications.

KEY FEATURES

- Custom designed PCB for ESP32 module.
- Compact and organized component layout.
- Proper power supply connections for stable operation.
- Accessible input and output pins for easy interfacing.
- Suitable for embedded and IoT projects.

ARCHITECTURE

- ESP32 module acts as the main controller.
- Power supply section provides required voltage to the ESP32.
- Input and output pins are connected through PCB tracks.
- Programming interface is provided for uploading code.
- All components are mounted on a single PCB.

RESULTS

- he ESP32 custom PCB was successfully designed and developed.
- All components were properly mounted and connected.
- The board powered on and functioned correctly.
- The project improved understanding of PCB design and ESP32 hardware.

FUTURE SCOPE

1. **The board can be improved with better power regulation.**
2. **The design can be made more compact and efficient.**
3. **The board can be improved with better power regulation.**